eedi-mining-misconceptions-in-mathematics

# Private 7th (Public 2nd) Place Solution Summary

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|---|---|
| **Rank** | 7 |
| **Team** | terekaerumasahmet |
| **Author** | kaerururu |
| **Topic ID** | 551388 |
| **Version** | Original |

Source: https://www.kaggle.com/competitions/eedi-mining-misconceptions-in-mathematics/discussion/551388
Retrieved with Kaggle CLI on 2026-10-21.

First of all, I'd like to thank kaggle team, host for hosting this competition , and my team mates (@tereka @irrohas @aerdem4)
I finally got 5th gold medal. I'm really happy to be Competitions Grand Master.

# Overview

Voting Ensemble of 3pipelines (Retriever + Reranker)
All reranker inference with vllm

# Pipeline1 by Masaya

This is brief summary.
Masaya will write up detail on this link.

Retriever

- Generate test answer with pretrained qwen35B
- model :SFR-Embedding-2_R, Hard Negative Sampling(N=100) removed if unseen in training data.
- dataset: Filled unknown Misconception with GPT4o-mini
- candidate 60

Reranker

- Qwen2.5 32B AWQ+LoRA (with 40 options)
  - options → 0~9 + A~Z + a~z (40 charactors)
- dataset: only original
- candidate 40
- run twice and merge ([:40] and [20:])
- left 25 with logprobs

# Pipeline2 by Tereka+Kaeru

Retriever

- model1 :SFR-Embedding-2_R、Hard Negative Sampling(100)
  - dataset: original + synthesis with gemma 27B/Qwen2.5 32B
  - lb 0.378
- model2 : public model by sayoulala
- weighted avg model1 and model2
- candidate 70 and filter with masaya preds
- only single fold

Reranker

- Qwen2.5 32B AWQ+LoRA (with binary options)
  - options → yes, not
- dataset: original + synthesis unknown misconceptions with gemma2 27B/Qwen2.5 32B
- only single fold

# Pipeline3 by Ahmet

Retriever

- Omitted in final pipeline

Reranker

- Candidates are each top9 from previous pipeline
- Qwen2.5 32B (with 9 options)

# Ensembling

- Voting ensemble of each pipeline
- Apply rank like following, join, sum rank, sort by sum_rank

```
sub0
  .with_columns(pl.col()..split())
  .with_columns(pl.lit([/i  i  ((, , ))]).alias())
  .explode(pl.col([, ]))
```

```
submission
  .sort(by=[, , , ], descending=[, , , ])
  .group_by([], maintain_order=)
  .agg(pl.col())
```

# Postprocessing

- Boost missing misconception rank

# Accelerating Tips

Retreiver

- Run parallel with cuda:0 and cuda:1
- Pre-calculating Misconception Vector
- Dynamic padding with collate_fn

Reranker

- Inference with vllm (AWQ 4bit quantization)
- Set vllm parameter enabling_prefix_cache True
- Filter candidate with previous preds of pipeline

## What not worked

Retrieval

- Concat vectors
- Avaraging many vectors

Reranker

- Full-data trained model
- QwQ model

Thank you!!